

$$3^2 + 4^2 = 9 + 16$$

ex 2.2

Find the derivative of $f(x,y) = xe^y + \cos(xy)$ at the point $(2,0)$ in the direction of $v = 3i - 4j$

^{sol}
Recall the direction of a vector v is the unit vector $\hat{v}$ is obtained by dividing v by its length. $a=3$ $b=-4$

$$u = \frac{v}{|v|} = \frac{3i - 4j}{\sqrt{3^2 + (-4)^2}} = \frac{3i - 4j}{\sqrt{9+16}} = \frac{3i - 4j}{\sqrt{25}} = \frac{3i - 4j}{5}$$

so calculate the unit vector $\hat{v} = \frac{v}{|v|} = \frac{3i - 4j}{5}$

step 1. $\hat{v} = \frac{3}{5}i - \frac{4}{5}j$

* The partial derivative of f are everywhere continuous and at $(2,0)$, are given by.

* $f(2,0) = (2)$

steps Find the gradient vector (∇f)

The gradient vector, ∇f , tells us the direction of the steepest ascent (greatest rate of increase) of the function. It is made up of the partial derivatives of f .

$$\nabla f(x,y) = f_x(x,y)i + f_y(x,y)j$$

1. Find the partial derivative with respect to x (f_x) (treat y as a constant)

$$f_x = \frac{d}{dx} = (ne^y + \cos(ny)) = (1 \cdot e^y) + (-\sin(ny))$$

$$f_x = e^y - y \sin(ny)$$

2. Find the partial derivative with respect to y (f_y): (Treat n as a constant)

$$f_y = \frac{d}{dy} = (ne^y + \cos(ny)) = (n \cdot e^y) + (-\sin(ny))$$

$$f_y = ne^y - n \sin(ny)$$

Steps: Evaluate the Gradient at the point $(2, 0)$

Now substitute $n=2$ and $y=0$ into the partial derivatives you found in Step 2.

1. Evaluate $f_x(2, 0)$

$$f_x(2, 0) = e^0 - (0) \sin(2 \cdot 0) = 1 - 0 = 1$$

This is $f_x(2, 0) = 1$ in the text.

2 Evaluate

$$f_y(2,0) = (2)e^0 - (2) \sin(2 \cdot 0) = 2 \cdot 1 - 0 = 2$$

3 From the Gradient Vector at $(2,0)$:

$$\nabla f(2,0) = f_x(2,0)\mathbf{i} + f_y(2,0)\mathbf{j}$$
$$\nabla f(2,0) = 1\mathbf{i} + 2\mathbf{j}$$

step 4, calculate the Directional Derivative (Dot)

The dot product of $\mathbf{A} =$
Finally, we calculate the directional derivative
by taking the dot product of
gradient vector (from step 3) and unit
vector (from step 1)

The dot product of $\mathbf{A} = a_1\mathbf{i} + a_2\mathbf{j}$
and $\mathbf{B} = b_1\mathbf{i} + b_2\mathbf{j}$ is $a_1b_1 + a_2b_2$

$$D_{\mathbf{u}}f(2,0) = \nabla f(2,0) \cdot \mathbf{u}$$
$$D_{\mathbf{u}}f(2,0) = (1\mathbf{i} + 2\mathbf{j}) \cdot \left(\frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}\right)$$

i components together multiply, and j
component together.

$$D_{\mathbf{u}}f(2,0) = 1 \cdot \left(\frac{3}{5}\right) + 2 \cdot \left(-\frac{4}{5}\right)$$

$$D_{\mathbf{u}}f(2,0) = \frac{3}{5} - \frac{8}{5}$$

$$D_{\mathbf{u}}f(2,0) = \frac{3-8}{5} = \frac{-5}{5} = -1$$